

Research Paper

Process design and economic evaluation for methane and electricity production from Agave bagasse using different pretreatments

Irmene Ortíz^{a,*}, Idania Valdez-Vazquez^{b,*}, Abigail Hernández-Vázquez^c,
Roberto Olivares-Hernández^a, Julián Carrillo-Reyes^b, Felipe Alatríste-Mondragón^d,
Germán Buitrón^b, Elías Razo-Flores^d

^a Depto. Procesos y Tecnología. Universidad Autónoma Metropolitana-Cuajimalpa. Av. Vasco de Quiroga 4871, Col. Santa Fe. C.P. 05348, México City, Mexico

^b Unidad Académica Juriquilla, Instituto de Ingeniería, Universidad Nacional Autónoma de México, Blvd. Juriquilla 3001, Santiago de Querétaro, Querétaro 76230, Mexico

^c Posgrado en Ciencias Naturales e Ingeniería. Universidad Autónoma Metropolitana-Cuajimalpa. Av. Vasco de Quiroga 4871, Col. Santa Fe. C.P. 05348, México City, Mexico

^d Instituto Potosino de Investigación Científica y Tecnológica A.C., División de Ciencias Ambientales. Camino a La Presa San José 2055, Lomas 4a Sección, C.P. 78216. San Luis Potosí, SLP, México

ARTICLE INFO

Keywords:

Pretreatment
Anaerobic digestion
Biomass
Biomethane
Techno-economic analysis

ABSTRACT

Agave bagasse is an agricultural residue that has the potential to be used as feedstock biorefineries. However, its complex structure requires a pretreatment step to facilitate the accessibility to carbohydrates for its subsequent conversion. In this study, the capital and operative expenses (CAPEX and OPEX) were simulated for industrial plants processing 500 ton/d of Agave bagasse using different pretreatment platforms for methane production and power generation on experimental data. The scenarios consisted of i) biological, ii) enzymatic, iii) dilute acid, and iv) steam explosion pretreatments and a case-based scenario without pretreatment. All these scenarios included milling, pretreatment, anaerobic digestion, and cogeneration stages. The results showed that those pretreatments carried out under soft conditions (biological and enzymatic) decreased between 2.1 and 2.8 times CAPEX compared to the highest value obtained in the diluted acid and case-based scenarios. The steam explosion, a pretreatment carried out under severe conditions, was less effective reducing CAPEX by only 1.2 times compared to the highest value. Furthermore, compared to the case-base scenario, only the biological pretreatment reduced the OPEX by a factor of 2, whereas the other pretreatments resulted in OPEX that were 2.0 to 3.5 times higher. Among the utilities (cooling water, steam generation, and energy) the steam generation contributed 47% to 76% of the total OPEX. The energy produced (kW-h/yr) was approximately twice the energy consumed for all scenarios. A sensitivity analysis demonstrated the robustness of the platforms since fluctuations in the prices of feedstock and raw materials did not impact the total production cost of methane, except for biological and enzymatic pretreatments. For all platforms, the solid loading was the process parameter with the highest impact on the production cost of methane. Overall, the study showed that biological and enzymatic pretreatments had the best economic performance. However, low substrate conversion and the high cost of enzymes and equipment, respectively, remained the main challenges to address in future research.

1. Introduction

Agave is a plant genus of the family Agavaceae of 200 to 300 species, including species grown commercially. The dry biomass yield of Agave spp. reported annually ranges from < 1 to 34 Mgha/yr without irrigation, depending on species and location [1]. Therefore, its potential as biofuel production feedstock has been pointed out, using the complete

plant or residues produced by some productive chains such as alcoholic beverages and fibers. Some theoretical calculations have indicated that the global Agave crop area alone could provide 6.1 billion L of ethanol if Agave were reestablished as a bioenergy feedstock without causing changes in land use [1].

Agave is the raw material in the Tequila industry and the Agave bagasse is the main waste from the production of this distillate (with the

* Corresponding authors.

E-mail addresses: irmene@cua.uam.mx (I. Ortíz), ivaldezv@iingen.unam.mx (I. Valdez-Vazquez).

<https://doi.org/10.1016/j.enconman.2024.118319>

Received 20 December 2023; Received in revised form 2 March 2024; Accepted 13 March 2024

Available online 20 March 2024

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appellation of origin). In 2022, the tequila industry processed 2,610,800 ton of Agave, of which 40 % corresponds to Agave bagasse (dry basis) in Mexico [2]. This lignocellulosic biomass is a renewable source suitable for biofuel production, including ethanol, diesel, hydrogen, and methane [3–5]. Nevertheless, the recalcitrance of lignocellulosic biomass requires a pretreatment step to facilitate the accessibility to carbohydrates for subsequent conversion. The pretreatments include biological, chemical, and physicochemical processes. Agave bagasse has been reported as a particularly high recalcitrance biomass compared to other lignocellulosic materials [6,7]. In a previous study, biological, enzymatic, chemical, and hydrothermal pretreatments applied to Agave bagasse were compared in terms of solubilization of carbohydrates (CHO) and chemical oxygen demand (COD), as well as the biochemical methane potential (BMP) from hydrolysates. An average yield of 0.16 ± 0.02 kgCOD/kg was reported for most pretreatments, except chemical pretreatment (where diluted acid was used), which value was 2.6 times higher [8]. However, the concentrations of inhibitors – furfural, hydroxymethylfurfural, and phenols – were more elevated in dilute acid hydrolysates than in the steam explosion while in the biological and enzymatic hydrolysates those compounds were not detected [8]. Further, the average BMP values from all types of hydrolysates were the same (219 ± 15 NL/kgCODin), indicating that the selection of the best pretreatment method for methane production must include not only the technical aspects but also the investment and operating costs analysis [8].

The production of biofuels at an industrial scale is a complex process, and as part of the design workflow for suitable production processes, the computational simulation allows the comparison of different configurations in terms of investment and operating costs, energy, water, and resource requirements, as well as, the environmental impact [9]. A recent study established the economic feasibility of bioethanol production from Agave in Mexico. Depending on the agricultural zone of the Agave, the production of ethanol could be profitable either for the farming sector or industrial sector depending on the factors considered, such as the costs of raw materials, the Agave culture, and the competition between mezcal and tequila industry [10]. Furthermore, another study on the technical and economic performance of bioethanol production from sugarcane and blue Agave bagasse indicated that the pretreatment (ozonolysis) had a minor effect on the ethanol costs and yield when compared to sugar cane bagasse because of its lower content of lignin and hemicellulose [11].

Nowadays, techno-economic analyses are valuable tools to compare the economic performance of emerging technologies and guide researchers to identify bottlenecks. In the case of methane production from Agave bagasse, the soluble organic matter recovered after pretreatment has been obtained under a wide range of parameters. For example, the diluted acid pretreatment was ca. 30 times faster in recovering twice as much the soluble organic matter than the biological treatment [8]. However, the diluted acid pretreatment resulted in a severity factor four times higher than the biological ones. In full-scale plants, the investment in expensive construction materials such as stainless-steel reactors, high operating costs, and energy consumption may make a project unprofitable [12]. Therefore, the objective of this study was to assess the techno-economic analysis of methane production and the power generation from Agave bagasse by comparing different pretreatment scenarios: i) biological, ii) enzymatic, iii) dilute acid, and iv) steam explosion, and a case-based with no pretreatment. The industrial plants include milling, pretreatment, anaerobic digestion, and cogeneration for all these scenarios. The main parameters evaluated were the total production costs of electricity per ton of Agave bagasse (kW-h/ton dry bagasse) and the energy efficiency. For the pretreatment and the anaerobic digestion stages, experimental data were used as parameters to perform the computational simulations of the different process configurations.

2. Materials and methods

The production of methane and power generation from Agave bagasse was modeled and evaluated using SuperPro Designer v11.2 (Intelligent Inc., Scotch Plains, NJ, USA) in a plant configuration that included four main sections a) biomass conditioning, b) pretreatment, c) methane production, and d) power generation, as shown in Fig. 1.

2.1. Biomass composition and conditioning

Considering the national availability [3], all scenarios received 500 ton/d of Agave bagasse. The biomass composition per ton was 564 kg of cellulose, 109 kg of hemicellulose, 152 kg of lignin, 35 kg of ash, other solids 74 kg, and 66 kg of water [8]. Biomass conditioning involved two rounds of shredding and grinding to reduce the size of particles. This was followed by screening to bring a final size of 3 ± 0.5 mm. The resulting ground biomass was then sent to a tank where it is hydrated and mechanically conveyed to the pretreatment section, as shown in Fig. 1 [13,14]. This section is highly power-consuming and represented between 77 % and 80 % of the total power demand for the different scenarios (Table 1S, Supplementary data).

2.2. Pretreatments and case-based scenarios

Five scenarios were contrasted, which differed in the pretreatment applied to the lignocellulosic biomass: biological, enzymatic, dilute acid, steam explosion, and a case-based scenario with no pretreatment as described below and as shown in Fig. 2a-d. The complete diagrams used by the simulator are shown in Fig. 1S–5S in Supplementary data. The operating conditions and process configurations of the four pretreatments considered were chosen from a previous study due to their notable results for solubilizing Agave bagasse and the operating conditions of the pretreatments were optimized separately in previous studies [8].

The biological pretreatment (Fig. 2a) was based on the joint action of the native microbiota of Agave bagasse composed mainly of *Clostridium*, *Lactobacillus*, and *Enterobacter*, described as a core microbiome of lignocellulosic biomasses [15]. This core microbiome primarily hydrolyzed xylan, converting it into volatile fatty acids, which were simulated solely as acetic acid to simplify the study. The enzymatic pretreatment (Fig. 2b) was based on the action of the commercial enzyme cocktail, Cellulase 50XL [16]. This enzyme cocktail was simulated to a dose of 293 endoglucanase units (EGU)/g of bagasse working in a 50 mM citrate buffer pH 6. This enzyme preparation released mainly glucose along with galactose, arabinose, and other sugars. These sugars represented 69 % of the total chemical oxygen demand (COD) simulated for this pretreatment [16]. The dilute acid pretreatment (Fig. 2c) solubilized all xylan and partially cellulose in 1.8 % HCl under soft operating conditions, which recovered sugars and eliminated the generation of inhibitors [17]. The steam explosion pretreatment (Fig. 2d) solubilized xylan producing a mixture of sugars and acetic acid [4]. The pretreatment was performed in a plug flow reactor coupled to a flask tank where the instant depressurization takes place. This pretreatment generated low concentrations of furfural (0.7 g/L) and hydroxymethylfurfural (0.12 g/L). The base scenario did not use any pretreatment unit and Agave bagasse was digested directly after milling and the digestate was used to produce power [8]. The general operating conditions for all pretreatments and the base scenario are listed in Table 1.

2.3. Methane production

After the pretreatment, the hydrolysates obtained were conducted to the anaerobic digestion stage, where part of the soluble organic matter was converted into methane and CO₂ in an Upflow Anaerobic Sludge Blanket (UASB) reactor. Each scenario considered experimental data of methane yields and COD removals according to studies that previously

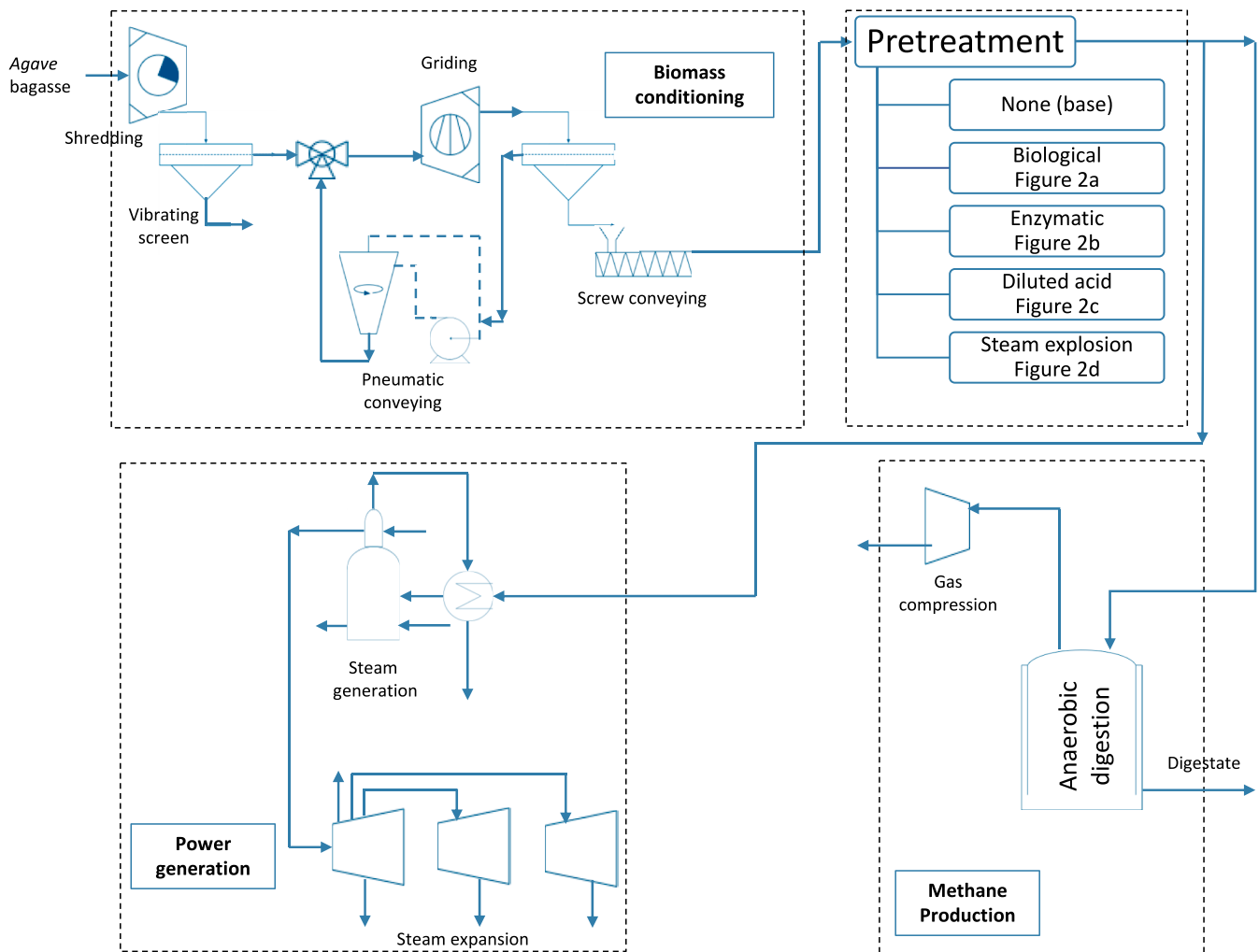


Fig. 1. General design of the plant for methane and power generation from Agave bagasse.

optimized the operating conditions (hydrolysate concentration and hydraulic retention time) for methane production from these hydrolysates [15–18]. Consequently, the simulated operating conditions did not consider the acclimatation stages; instead, methane yields and COD removals corresponded to the stable period of bioreactors. The methane generation was simulated using a set of anaerobic digestors (Fig. 1). The general operating conditions for the anaerobic digestion stage in all scenarios are listed in Table 1.

2.4. Power generation

Steam and electrical power were generated in the cogeneration stage by burning the unconverted solids remaining from the pretreatment stage. The properties and composition of these streams are presented in Table 2S in Supplementary data. The main components of this solid streams are cellulose and lignine ranging from 44% to 65 % and 17–24 %, respectively. In the case-based scenario, the electricity was produced from digestate of the anaerobic digestion stage. In this section, a steam generator produces high-pressure steam to feed the steam turbine generator that produces the electricity. The physical properties were estimated based on the database available in SuperPro Designer considering the solid composition. These stages are shown in Fig. 1 [20].

2.5. Economic assessment

The capital expenses (CAPEX) and operative expenses (OPEX) for

each bioprocess configuration were calculated once the mass balances were solved; both were obtained using Super Pro Designer®. The prices of the raw materials and the related costs used for calculations are listed in Table 2. The main goal of the economic evaluation for these processes was to relate the energy and methane production for each configuration to CAPEX and OPEX, and the identification of the key factors that impact the economic feasibility of the Agave bagasse-based plant.

2.6. Sensitivity analysis

A sensitivity analysis was conducted to assess the effects of changes in process parameters and raw material prices using the methane production cost (USD/kg) as the response variable. To ensure a fair comparison between the pretreatment platforms, operating conditions, and methane yield were standardized. Specifically, a residence time of 76 h and 221 L CH₄/kg COD_{in} was used across all pretreatment platforms [8]. Two groups of values were modified to evaluate their impact on the production cost of methane. The first group pertained to process parameters such as total solids (%), residence time (h), and temperature (°C), which were varied by ± 50 % in comparison to the values listed in Table 1 [21]. However, for steam explosion and diluted acid pretreatments, the selected temperatures were 116 °C and 100 °C for the low levels, respectively, and 152 °C for the high level since steam was used as the heat transfer agent. The second group of values pertained to raw materials, including Agave bagasse, water, and sodium hydroxide listed in Table 2, which price, was also varied by ± 50 %. These prices

Table 1

Operating conditions for the stages of pretreatment and anaerobic digestion simulated for each scenario.

Parameter	Case-based	Biological	Enzymatic hydrolysis	Dilute acid	Steam explosion
Pretreatment stage					
Solids loading (% wt/wt)	–	10.0	8.8	5.0	2.7
COD _{out} (g/L)	–	20.3	41.1	20.7	10.2
Temperature (°C)	–	37.0	42.7	119.0	154.0
Residence time (h)	–	60.0	2.5	1.7	0.2
Reactor material	–	concrete	stainless steel	stainless steel	stainless steel
Catalyst	–	Acidogenic bacteria	Cellulase 50XL; 6.1 kg/ton Agave	1.8 % (v/v) HCl	H ₂ O
Total volatile solids conversion (%)		16	27	35	13
Reference		[15]	[16]	[17]	[4]
Anaerobic digestion stage					
COD _{in} (g/L)	10 ^a	20.3	25.0	10.0	10.2
Hydraulic retention time (h)	480	80	30	72	168
COD removal %	33	76	94	80	89
Yield (LCH ₄ /kg COD _{in})	128	220	233	280	290
Reference	[8]	[19]	[16]	[17]	[18]

a In the case-based scenario, the Agave bagasse is fed directly to the reactor at a total solid content of 10 g/L.

were common to all pretreatments. In addition, the costs of hydrochloric acid (HCl) and enzymes, which are specific to the diluted acid and enzymatic pretreatments, respectively, were also varied by $\pm 50\%$.

3. Results and discussion

The main results of the simulations, including overall balances by component used for the technical and economic analysis presented in this section are included as Supplemental Information (Tables 3S to 17S).

3.1. Methane production

Due to the nature of each pretreatment, hydrolysates differed in composition: volatile fatty acids in the case of the biological pretreatment, a mixture of hexoses in the case of the enzymatic pretreatment, a

Table 2

Prices for the raw material across the different scenarios.

Raw material	Price	Unit
Agave bagasse	0.05031	USD/kg
Sodium hydroxide (NaOH) ^a	1.114	USD/kg
Cellulase enzyme ^b	190	USD/kg
Citric acid (C ₆ H ₈ O ₇)	0.71	USD/kg
Sodium citrate (Na ₃ C ₆ H ₅ O ₇)	0.9	USD/kg
Hydrochloric acid (HCl)	0.317	USD/kg
Water ^c	0.45	USD/m ³

a. Average price based on 2023. b. Cellulase from *Aspergillus niger* with ~ 0.8 U/mg. c. Water for industrial purposes in the state of Jalisco (CONAGUA, Mexico). The rest of the costs of the different raw materials are established in the SuperPro databases.

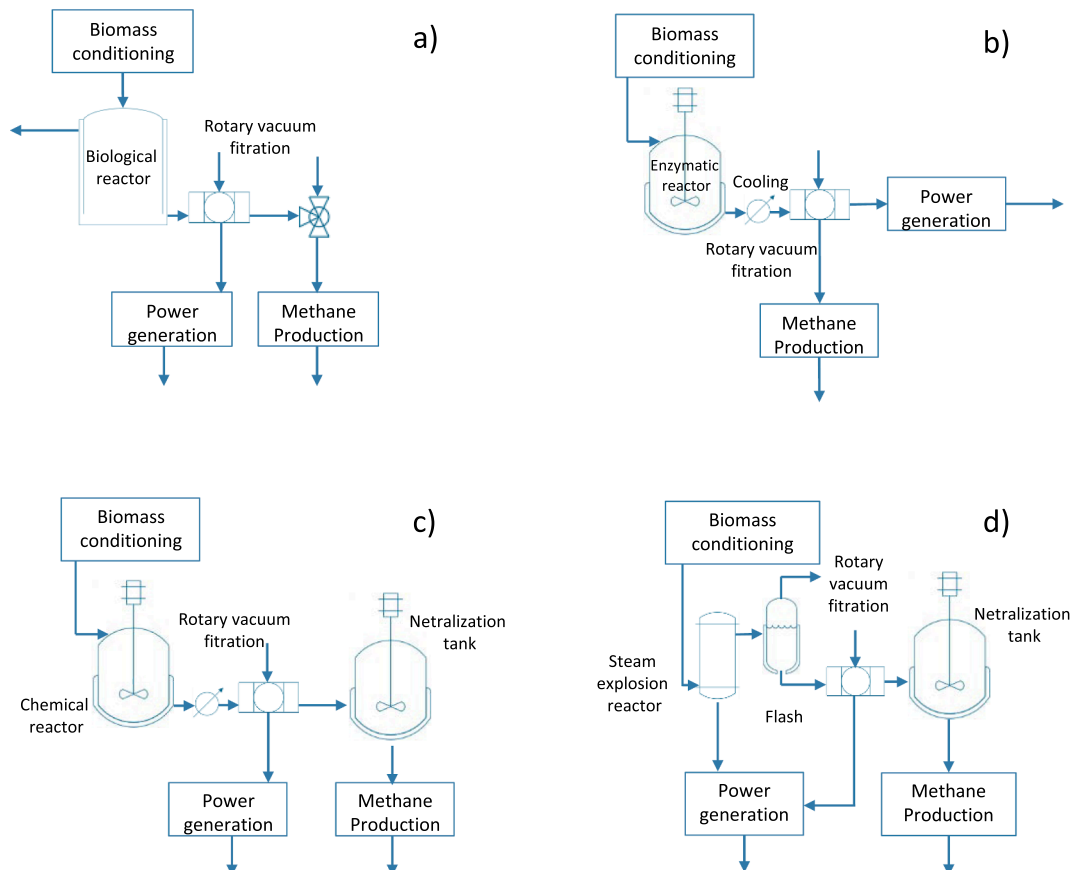


Fig. 2. Pretreatments design scenarios. a) Biological. b) Enzymatic. c) Diluted acid. d) Steam explosion.

Table 3

Methane produced in each pretreatment at 37 °C, 1 atm, processing 500 ton bagasse/day.

Scenario	Mass Methane Production (ton/yr)	Volumetric Methane Production (m ³ /yr)
Case-based	14,618.87	46,381,454
Biological	4,509.54	16,602,962
Enzymatic	12,314.03	49,663,924
Diluted acid	11,795.35	37,810,066
Steam explosion	4,513.95	14,442,586

mixture of pentoses, acetic acid, and soluble lignin in the case of the steam explosion pretreatment, and a combination of hexoses and pentoses in the case of the dilute acid pretreatment [8]. Based on this data, it was possible to compare the amount of methane produced per year, as shown in Table 3. The methane production depended primarily on the performance of the pretreatment stages, i.e., the solid loading and solubilization efficiency. The methane yields were very similar among scenarios since the differences are only by 13 % (Table 1). The case-based scenario considered that the Agave bagasse was directly methanized after milling. The methane digester reached a removal of total solids of 33 % with a methane production of $464 \times 10^5 \text{ m}^3/\text{yr}$ at the expense of the largest hydraulic retention time of the scenarios studied. Then, the enzymatic hydrolysis was the best scenario for methane production since it reached the highest solubilization among scenarios, 12 ton/d (obtained by multiplying the total volatile solids conversion by the mass flow), of which 94 % were converted into methane. The enzymatic pretreatment unit was one of the scenarios that operated at a high-solid loading combined with a high percentage of solubilization. In the second place, the diluted acid pretreatment solubilized 9 ton/d with a percentage of conversion to methane 14 % lower than the enzymatic pretreatment. The diluted acid pretreatment had the highest percentage of solubilization but operated at a low-solid loading. The biological and steam explosion pretreatments resulted in similar methane production, but the operating conditions largely differed. The biological pretreatment operated at the highest solids loading among scenarios but with a low solubilization percentage, resulting in 8 ton/d of bagasse solubilized. In contrast, the steam explosion pretreatment operated at the lowest solids loading among scenarios with a low percentage of solubilization that resulted in just 2 ton/d of bagasse solubilized.

This study showed that methane production depended mainly on the efficiency of the pretreatment unit and not on the methane digester. The two bottlenecks identified in the pretreatment stage were the solid loadings and the solubilization efficiency. As the solid loadings increase, the rheological behavior and transport phenomena are negatively affected in the reactors [22]. The pretreatment sections presented in this study showed that the solubilization percentage peaked at a solid loading of 5 % and then decreased; this behavior could be attributed to mass transfer limitations as the solid loadings increased, as previously reported [23]. Novel bioreactor technologies must consider operating at high solids loading – ideally higher than 15 % – and ensure good solubilization. Regarding the solubilization percentage, values from 13 % to 34 % were observed in the studied scenarios. The solubilization percentage depends on the type of pretreatment since each one has a different action over the lignocellulosic material. For example, the biological and steam explosion pretreatments only target the xylan fraction; consequently, the maximum solubilization percentage largely depends on the xylan content in the feedstock. The biological pretreatment reached 100 % solubilization of the xylan fraction; therefore, a higher solubilization percentage requires the action of microorganisms that hydrolyze cellulose, which undoubtedly requires increasing the hydraulic retention time in the pretreatment section. In comparison, the steam explosion pretreatment reached 55 % solubilization of the xylan fraction. In contrast, the dilute acid and enzymatic pretreatments

solubilized ca. 50 % of cellulose – the majority fraction of feedstock – outperforming other pretreatments in efficiency.

3.2. Power generation

For each scenario, energy generation was evaluated using a set of steam turbines. The steam was generated by burning the solids separated after the pretreatments. The consideration of burning only methane for steam generation resulted in an unfavorable energy balance between the energy consumed in the process and the energy produced by the set of turbines. An interesting scenario to explore could be the enzymatic hydrolyzation of the solids after pretreatment and its posterior conversion to methane or other valuable products in a biorefinery scheme. As discussed earlier, the effectiveness of organic matter solubility in some pretreatments was limited and remained in the solids. Notably, the solids resulting after the steam explosion pretreatment are susceptible to enzymatic hydrolysis as reported, with a biochemical methane potential (BMP) of hydrolysates (pretreated at 142 °C-15 min and hydrolysed by Cellic CTec2) of 284 ± 20 in NL CH₄/kg_{COD} with a COD removal of 78.4 ± 1.3 [24].

On the contrary, the net energy balances for power generation systems were favorable for all the scenarios; the energy produced (kW·h/yr) was approximately two times the energy consumed as shown in Fig. 3.

3.3. Economic assessment

The size and equipment needed for processing 500 ton/day of Agave bagasse were fitted according to the mass and energy balances for each scenario, and the CAPEX and OPEX simulated are shown in Fig. 4. The highest CAPEX was for the diluted acid pretreatment scenario; the cost was related to the size and materials needed for the reactors to resist the acidic conditions of the process (Fig. 4). In contrast, the highest OPEXs were for the steam explosion and diluted acid pretreatment scenarios related to the energy requirements (heating, steam generation, and expansion).

For further analysis, the OPEX was broken down into three categories: the raw material, utilities, and facility-dependent costs; their contribution to the total OPEX is presented in Table 4. For the case-base, biological, and diluted scenarios, the main contribution to the OPEX is

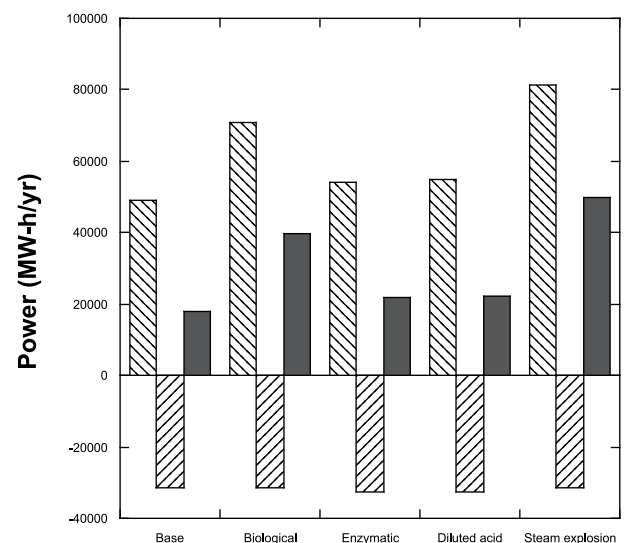


Fig. 3. Power generated (hatched) and consumed (diagonal lines) and net energy (solid) for each pretreatment in kW·h/yr processing 500 ton bagasse/day.

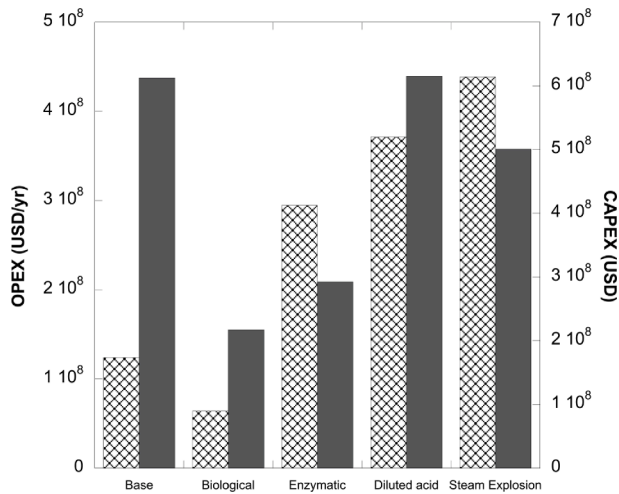


Fig. 4. Total CAPEX (■) and OPEX (▨) for the different scenarios.

Table 4

Main factors affecting the annual OPEX. The percentage represents the participation of the factor in the total OPEX.

	Base	Biological	Enzymatic	Diluted acid	Steam explosion
Raw materials	6.78	28.37	79.81	22.46	3.03
Facility-Dependent	88.03	60.93	17.18	29.03	19.42
Utilities	3.52	6.15	1.84	47.34	76.47

facility-dependent (depreciation of the fixed capital investment and maintenance of the facility). In contrast, for the enzymatic scenario, as expected, the main contribution was the raw material related to the cost of the cellulase enzyme. In the steam explosion and diluted acid scenarios, the utilities represent the highest costs, including cooling water, steam generation, and energy. On the other hand, the highest OPEX related to the raw materials corresponded to the enzymatic pretreatment scenario since the enzyme was the highest price (Table 2), and

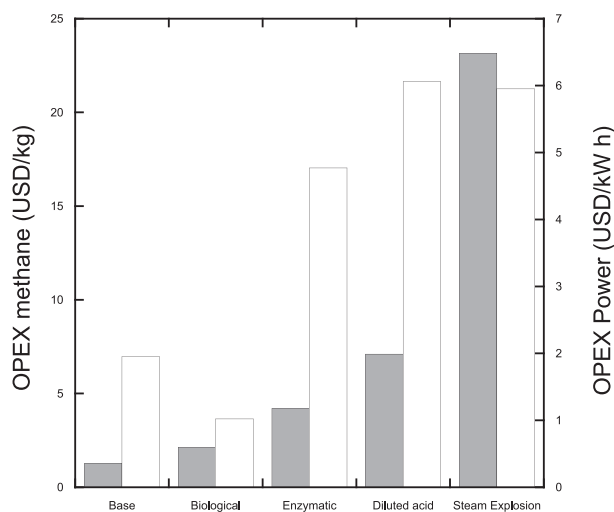


Fig. 5. OPEX related to production of methane (■); and power (▨) for each scenario processing 500 ton bagasse/day.

approximately 966 ton/yr of cellulase is needed.

A relative production cost analysis was calculated, considering the amount of methane and energy balance (Fig. 5). The lowest price for energy generation was for the biological pretreatment, 1.02 USD/kW · h, and the lowest cost for producing methane was 1.26 USD/kg obtained in the case-based scenario. The steam explosion had the highest operating cost for methane and power production as shown in Fig. 5, mainly related to the cost of utilities.

According to the sustainable development goals, access to clean and affordable energy must be ensured. Therefore, renewable energy production processes and diversifying energy sources are required. In Mexico, the production of primary energy comes mainly from natural gas (49.11 %) and crude oil (36.30 %), the rest from renewable resources, uranium, and coal, according to the Mexican Ministry of Energy [25]. Further, 48.59 % of primary energy generated worldwide comes from crude oil [26].

Based on the net energy balance of the designed scenarios, the equivalent in terms of barrels of crude oil was calculated and shown in Table 5, considering that the annual average of kW · h generated for crude oil was 532.98 kW · h /barrel [27]. The equivalent of energy in crude oil barrels produced in the steam explosion scenario was the highest, 15.7 % and 20.2 % higher compared to case-based and diluted acid scenarios.

In comparison, another system that used lignocellulosic material in biofuels had similar observations regarding the economic and environmental feasibility. For instance, converting corn stover thermochemical routes of gasification and mixed alcohols (GMA) system has shown environmental advantages but is not economically feasible. In contrast, the acetone–butanol–ethanol (ABE system) process is an economically and environmentally viable route requiring further improvement and detailed process engineering [28].

3.4. Sensitivity analysis

The Fig. 6 shows the percentage deviations relative to the cost of methane production (presented in section 3.3). The sign of the percentage represents either an increment (+) or a decrement (-) of that cost. The impact of solid loading on the total production cost of methane was the highest, as shown in Fig. 6a, consistent with the previous discussion in section 3.1. For instance, increasing the total solids from 2.7 % to 5.4 % during steam explosion pretreatment, reduced the total production cost of methane by 61 % (Fig. 6a). In the case of diluted acid and biological pretreatments, increasing the total solids from 5 % to 10 % and 10 % to 20 %, reduced the total production cost by 22 %. The enzymatic pretreatment, however, was less sensitive to the increase in solid loading. When the solid loading was decreased by 50 %, all pretreatment platforms experienced an increase in the total production cost, ranging from 9 % to 30 % due to the increase in the reactor size.

As for the residence time, reducing it by 50 % marginally reduced the methane production costs from 3 % to 6 %. Further, the steam explosion pretreatment was insensitive to the change in residence time. Conversely, the biological pretreatment had the highest increase of 16 % when the residence time increased from 60 h to 120 h. On the other hand, the temperature reduction had the highest impact on the methane production cost for the diluted acid and steam explosion pretreatments.

Table 5

The equivalent of crude oil barrels produced for each scenario.

Scenario	barrels/yr
Base	31,615.08
Biological	70,565.43
Enzymatic	38,610.56
Diluted acid	39,734.57
Steam explosion	88,875.17

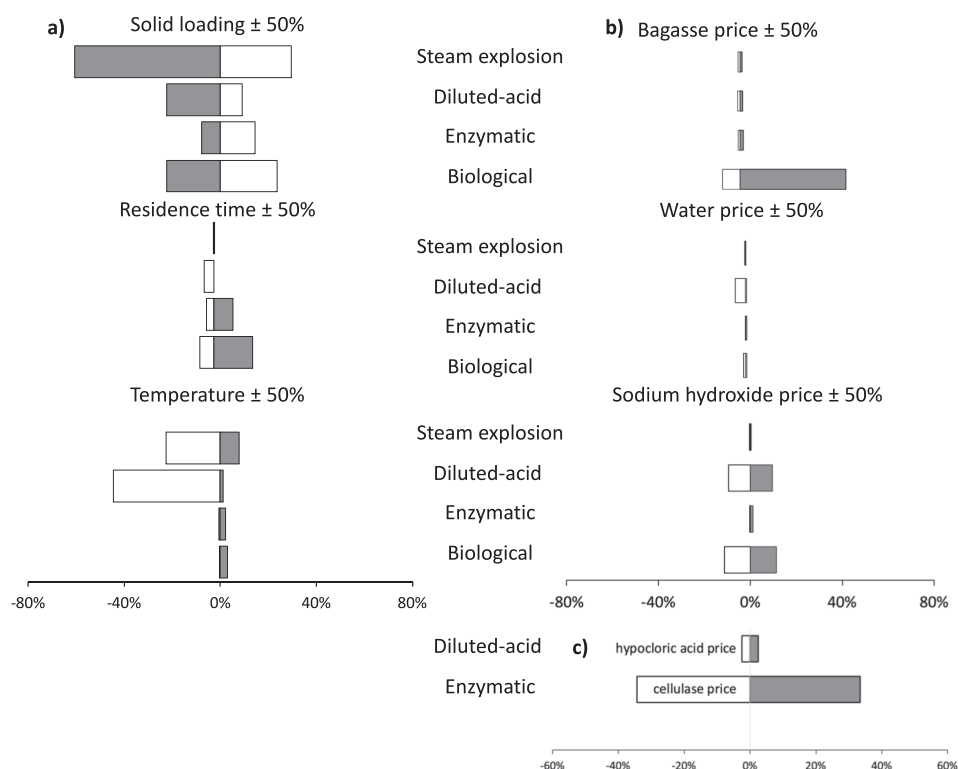


Fig. 6. Methane production cost variation (USD/kg) for the different scenarios, as a function of variations of a) Operation parameters, b) Common raw materials prices, and, c) Specific raw materials prices for enzymatic and diluted acid scenarios. Open symbols -50% and filled symbols $+50\%$ variation on the parameters or prices.

The temperature reduction from $119\text{ }^{\circ}\text{C}$ to $100\text{ }^{\circ}\text{C}$ in the diluted acid pretreatment and from $154\text{ }^{\circ}\text{C}$ to $116\text{ }^{\circ}\text{C}$ in the steam explosion pretreatment reduced the production cost by 45% and 23% , respectively.

Fig. 6b presents the sensitivity analysis for common raw materials for all the pretreatments which include *Agave* bagasse, water, and sodium hydroxide. The cost of methane increased by approximately 52% when the price of *Agave* bagasse was increased in the biological pretreatment scenario. This was due to the low amount of methane production in the biological pretreatment. On the other hand, for the rest of the pretreatments, the low response (less than 2%) to the prices of *Agave* bagasse can be attributed to the low process costs related to this particular raw material as compared to utilities like steam or energy. Similarly, the water price variation resulted in a low response, which was the same for all the pretreatments. In the biological and diluted acid pretreatment scenarios, the price of sodium hydroxide had a 10% response, which was related to the larger amount of this compound in those pretreatments compared to the explosion or enzymatic pretreatment scenarios.

In general, the production cost of methane was not significantly affected by the variations in common raw material costs evaluated because other factors, such as facility-dependent and utility costs, contributed to about 80% of the annual OPEX (Table 4). However, the price of the enzyme used in enzymatic pretreatment had an impact of 33% in both directions (Fig. 6c). The complete data on the cost variation analysis is presented in Table 18S as Supplementary data. These results indicated that optimizing each pretreatment independently is necessary, as several factors have an impact on the process and the production cost of methane does not have a linear response to the parameters and raw material prices.

4. Conclusions

This study showed that the pretreatment influenced mainly the

CAPEX and, to a lesser extent, the OPEX of lignocellulosic plants processing *Agave* bagasse. The biological pretreatment was the most economical scenario, whereas the steam explosion pretreatment was the most expensive. The two bottlenecks identified in the pretreatment stage were the solid loadings and the solubilization efficiency. The consideration of burning only methane for steam generation resulted in an unfavorable energy balance between the energy consumed in the process and the energy produced by the set of turbines. On the contrary, the net energy balances for power generation systems were favorable for all the scenarios; the energy produced ($\text{kW} \cdot \text{h} / \text{yr}$) was approximately two times the energy consumed. The sensitivity analysis showed that the solid loading had the highest impact on the total production cost of methane. At the same time, the prices of raw materials affected the total production cost to a lesser extent, except for the costs of enzymes and feedstock that impacted the enzymatic and biological pretreatments, respectively.

This work was financed by the Mexican Department of Energy (SENER) through the Gaseous Biofuels Cluster Project 247006.

CRediT authorship contribution statement

Irmene Ortíz: Writing – original draft, Visualization, Conceptualization. **Idania Valdez-Vazquez:** Writing – original draft, Software, Conceptualization. **Abigail Hernández-Vázquez:** Software, Investigation. **Roberto Olivares-Hernández:** Writing – original draft, Software. **Julián Carrillo-Reyes:** Writing – review & editing, Investigation. **Felipe Alatraste-Mondragón:** Investigation. **Germán Buitrón:** Writing – review & editing, Supervision, Funding acquisition. **Elías Razo-Flores:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enconman.2024.118319>.

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